

Supplementary Information for The Spatial Organization of Cold and Warm Temperature Synchrony

ASymClim Working Group

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Part I

Data and primary synchrony definitions

S1 Detailed PRISM data access and preprocessing

The analysis uses daily minimum and maximum temperature from PRISM. PRISM combines station observations with physiographic interpolation to represent temperature and precipitation over complex terrain [Daly et al., 2008]. Source access records provider, product, statistic, query bounds, coordinate reference information, retrieval mode, and source provenance. The scientific analysis must reject synthetic fallback.

The validated center-pixel walkthrough acquired observations for longitude -105.75 to -104.75 and latitude 39.50 to 40.50 from 1 November 2023 through 30 January 2024. The selected center was latitude 40.0 , longitude -105.25 . The bounded relationship analysis reused the corresponding observed winter data over a 20 by 20 pixel block. The precise source files, retrieval timestamps, checksums, and serving-revision records remain project evidence and must accompany the archived analysis bundle.

The broader center-pixel walkthrough also illustrated 547 rolling center maps over a 1981–2025 record. Those maps demonstrate the single-center workflow; they are not repeated full local-surface analyses and are not used as evidence for temporal persistence in the present technical report.

Preprocessing retains daily coordinate labels, native missing values, and the PRISM grid. The tail calculation performs pair-specific missing-value removal. It does not fill missing temperature values, interpolate through time, or standardize each series before ranking. Temperature units do not affect Spearman correlation, but units and source identity remain required provenance.

The Colorado analysis used the same 1 November 2023 through 30 January 2024 observed PRISM window and retained Colorado focal pixels with a 100-km local observation window. External tile halos prevented computational boundaries from truncating eligible relationships.

COLORADO PROVENANCE PLACEHOLDER: Insert final Colorado and halo bounding box, state-boundary source and version, PRISM grid resolution, retrieval manifests, checksums, serving revision, missingness summaries, and any source-side request partitioning.

S2 Mathematical definition of cold-tail synchrony

Let $X_{i,t}^{\min}$ and $X_{j,t}^{\min}$ be daily minimum temperature at pixels i and j . After removing times missing from either member of the pair, define the pixel-specific lower thresholds

$$q_i^{\min} = Q_b(X_i^{\min}), \quad q_j^{\min} = Q_b(X_j^{\min}), \quad (\text{S1})$$

where $0 < b \leq 0.5$ and $b = 0.5$ in the validated analysis. The joint cold set is

$$\mathcal{T}_{ij}^{\text{cold}} = \{t : X_{i,t}^{\min} \leq q_i^{\min}, X_{j,t}^{\min} \leq q_j^{\min}\}. \quad (\text{S2})$$

If $|\mathcal{T}_{ij}^{\text{cold}}| \geq \text{min_t}$ and both ranked series have nonzero variation, cold synchrony is

$$S_{\text{cold}}(i, j) = \text{cor} \left[\text{rank}(X_{i,t}^{\text{min}}), \text{rank}(X_{j,t}^{\text{min}}) \right]_{t \in \mathcal{T}_{ij}^{\text{cold}}} . \quad (\text{S3})$$

Otherwise it is undefined.

S3 Mathematical definition of warm-tail synchrony

Let $X_{i,t}^{\text{max}}$ and $X_{j,t}^{\text{max}}$ be daily maximum temperature. Define the upper thresholds

$$q_i^{\text{max}} = Q_{1-b}(X_i^{\text{max}}), \quad q_j^{\text{max}} = Q_{1-b}(X_j^{\text{max}}). \quad (\text{S4})$$

The joint warm set is

$$\mathcal{T}_{ij}^{\text{warm}} = \{t : X_{i,t}^{\text{max}} > q_i^{\text{max}}, X_{j,t}^{\text{max}} > q_j^{\text{max}}\}. \quad (\text{S5})$$

The strict upper-tail inequality is part of the validated implementation. Warm synchrony is the average-rank Spearman correlation evaluated on that set when its count meets `min_t` and the denominator is nonzero.

S4 Cold-warm asymmetry

The pairwise contrast is

$$\Delta S(i, j) = S_{\text{cold}}(i, j) - S_{\text{warm}}(i, j). \quad (\text{S6})$$

Its valid numerical range is $[-2, 2]$. Positive values indicate stronger cold-tail synchrony and negative values indicate stronger warm-tail synchrony. Because cold and warm coefficients use different variables and masks, ΔS is not a correlation coefficient and should not be constrained to $[-1, 1]$.

The primary focal-pixel reduction is

$$D_p = \text{median}_c[\Delta S(c, p)]. \quad (\text{S7})$$

It is not the difference between the separately reduced component fields,

$$\text{median}_c[S_{\text{cold}}(c, p)] - \text{median}_c[S_{\text{warm}}(c, p)], \quad (\text{S8})$$

which need not equal D_p . Technical-report ΔS maps use red for negative values and stronger warm-tail synchrony, near-white for zero, and blue for positive values and stronger cold-tail synchrony.

S5 Window and threshold semantics

The calculation selects a trailing coordinate-label interval from `window_end-window_days` through `window_end`, inclusive. For complete daily data, `window_days=90` therefore contains 91 labels. The reported window is exactly 1 November 2023 through 30 January 2024. We call it the 91-day winter window while retaining the implementation parameter name. Thresholds are calculated independently for each series after pairwise missingness filtering and within the selected window. In a rolling analysis, the thresholds are recalculated for every output window. In the bounded

experiment, one selected window is materialized and all pairs use that temporal support. Because filtering is pair specific, a pixel’s threshold can differ among pairs when missingness patterns differ.

The term “rolling median” should therefore be read as a median recalculated within each moving window, not as a daily smoothed median time series. The statewide implementation must record every window end and must not infer record bounds from stale per-batch attributes after concatenation.

S6 Joint-tail masking

Tail membership is joint. A day contributes to cold synchrony only when both pixels are at or below their own lower thresholds. A day contributes to warm synchrony only when both are above their own upper thresholds. Selecting tail days at one location and correlating the unconditioned values at the other would define a different statistic. So would using a regional threshold shared by all pixels.

The implementation records `cold_joint_count` and `warm_joint_count` for every relationship. These counts should appear in quality-control maps and figure source data. A valid coefficient with a small joint count should not be treated as equally stable to one with a larger count.

S7 Spearman implementation

Spearman correlation is computed as the Pearson correlation of ranks [Spearman, 1904]. For selected values x_k and y_k , let r_k and s_k be their ranks. Then

$$\rho_S = \frac{\sum_k (r_k - \bar{r})(s_k - \bar{s})}{\sqrt{\sum_k (r_k - \bar{r})^2 \sum_k (s_k - \bar{s})^2}}. \quad (\text{S9})$$

The implementation returns undefined if the denominator is zero or not finite. It does not substitute zero correlation for a constant selected series.

S8 Tie handling

Ranks are one based and tied values receive their average rank. A stable sort is used before tied blocks are identified. This choice matters for gridded daily temperature because quantization and repeated values can generate ties. Unit tests cover monotone values, tied values, opposed signals, independent signals, and finite range constraints.

S9 Missing data and minimum tail count

Missingness is handled before thresholds are calculated. For each variable and pair, times missing from either pixel are removed. If fewer than `min_t` paired observations remain, the coefficient is undefined and the joint count is zero. If the full paired record is sufficient but fewer than `min_t`

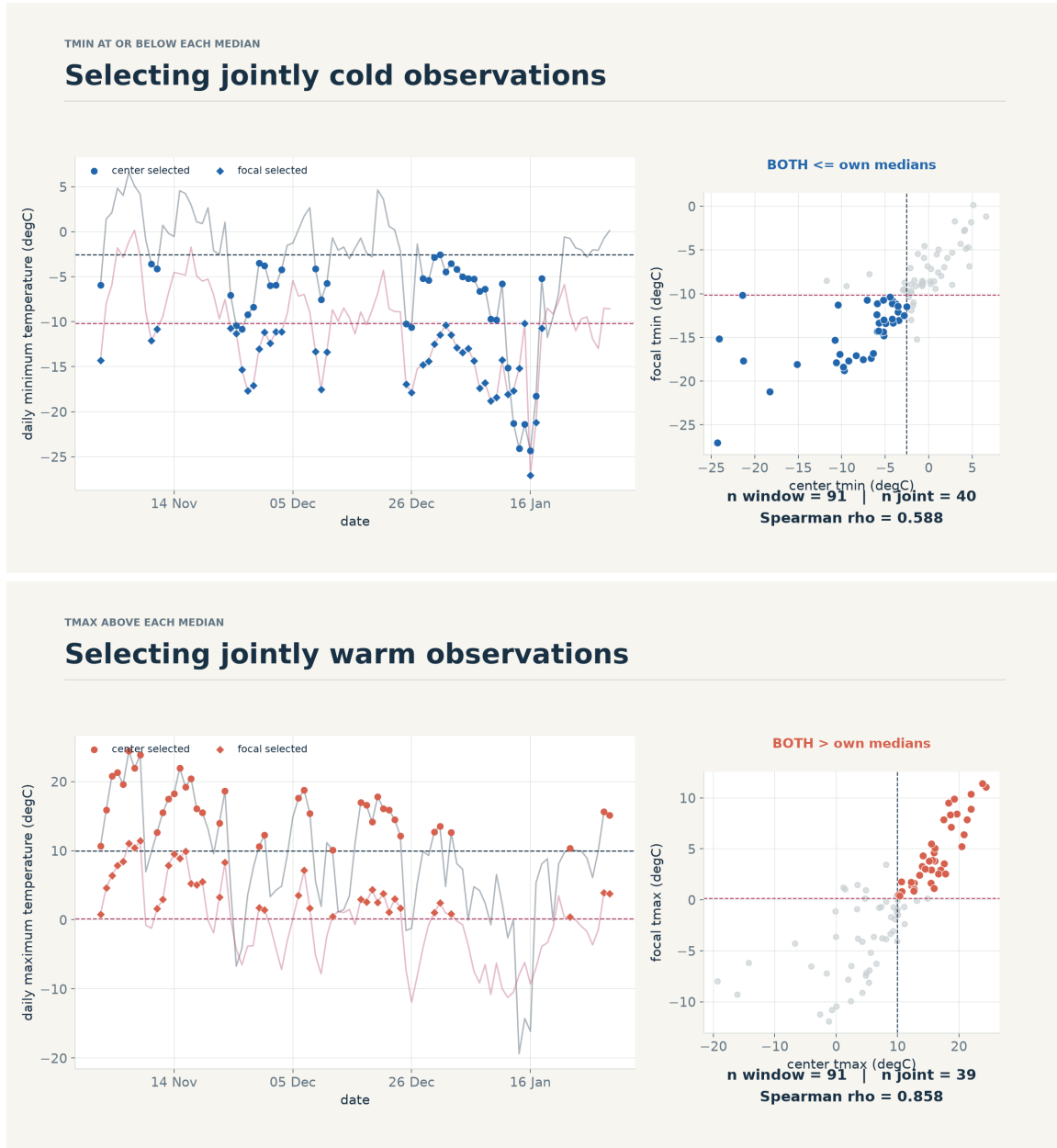


Figure S1: Audited joint-tail selection for one real PRISM pair. Cold synchrony uses days when both minimum-temperature series are at or below their own medians. Warm synchrony uses days when both maximum-temperature series are strictly above their own medians. The example retained 40 cold days and 39 warm days.

observations meet the joint-tail mask, the coefficient is undefined and the actual selected count is retained.

The bounded analyses used `min_t=10`. The audited focal pair contained 40 cold and 39 warm joint observations. The exact Colorado minimum count is not recorded in the retained report and must be recovered from the run manifest before archival release; the public software default is not a substitute for the analysis setting.

Part II

Validation and calculation semantics

S10 Pair symmetry

The implemented pair statistic is symmetric because paired missingness, per-series quantiles, the joint mask, and rank correlation are unchanged when i and j are exchanged. Thus

$$S_{\text{cold}}(i, j) = S_{\text{cold}}(j, i), \quad S_{\text{warm}}(i, j) = S_{\text{warm}}(j, i), \quad \Delta S(i, j) = \Delta S(j, i). \quad (\text{S10})$$

The calculation maps each directed request to a canonical pair $(\min(i, j), \max(i, j))$, computes each unique code once, and gathers the result back to both endpoint positions. Tests verify symmetry with missing values, ties, and insufficient tails. The stack-to-edge conversion can recheck equality to an absolute tolerance of 10^{-7} .

S11 End-to-end cold, warm, and Delta S semantics audit

A separate forensic audit traced observed PRISM temperatures through variable routing, pairwise missing-value removal, median thresholds, joint-tail masks, average ranks, pairwise coefficients, ΔS , focal reduction, NetCDF and GeoTIFF storage, and displayed color. Five observed relationships recomputed directly with SciPy agreed with retained stored values to a maximum absolute error of 2.22×10^{-16} . Three known-answer controls returned ΔS of +0.925, -0.925, and 0 for cold-stronger, warm-stronger, and equal cases, respectively. No numerical sign reversal was found.

For one western focal pixel, the audit independently recomputed 1,846 nonself relationships. Median cold synchrony was -0.025902, median warm synchrony was 0.881413, and the primary median of pairwise ΔS was -0.875348. The difference between the two component medians was -0.907315, a discrepancy of 0.031967 that confirms why the two reductions require distinct names. The primary value agreed exactly with the NetCDF output and to 2.55×10^{-9} with the float32 GeoTIFF.

The audited one-window CONUS product had national medians of 0.910874 for cold synchrony, 0.928470 for warm synchrony, and -0.017335 for median pairwise ΔS ; 70.5 percent of finite cells were negative. These values verify the output sign and storage semantics. They are not analyzed as a national scientific result in this technical report because temporal replication, national coverage review, and the relevant scientific quality controls remain separate requirements.

The reviewed high-resolution atlas contained 481,630 complete focal cells on a 1399 by 621 output grid, used a 100-km neighborhood, and excluded self-pairs. The audited PRISM input, CONUS

result, and semantics manifest had SHA-256 hashes 7ce255e9e09bebb824a6a7f6326a8ce4de18576148eaa837c89f2852c071092f, f3fbced095c324ba752a8d4a25c0b67da4e2935b44f534dcf30385b4f96d5709, and aa77b41105de70566711b51432ac9155718e5df3ba6ed5d6617d85ecab45b04d, respectively. These identifiers establish which files were audited; they do not substitute for complete source and configuration provenance.

The audit found communication inconsistencies rather than computational ones: one generic CONUS legend, historical figures using the opposite diverging palette, and mixed “hot” and “warm” terminology. This technical report uses “warm” and the shared red-negative, blue-positive contract. The affected historical panels used here were recolored without changing their values. The focused audit tests and full offline suite passed, with 980 tests passed, five skipped, and 419 deselected in the recorded run.

Part III

Spatial representation, support, and scale

S12 Center-reference implementation

A center-reference landscape is a two-dimensional slice of the stack at one center index. It retains cold synchrony, warm synchrony, ΔS , joint counts, distance, bearing, distance band, and direction code. The bounded equivalence audit compared these slices against the prior rolling center-reference method. Maximum absolute error was 0.0 for all three synchrony variables.

Self-pairs were retained in the bounded reductions. When a selected tail has sufficient variation, a self-pair has correlation one in both tails and ΔS zero. The validated bounded reductions summarized all finite centers and therefore included one self-pair per focal stack. The later fixed 100-km control and empirical-range pilot excluded self-pairs. The archived manifest must identify the rule used for every Colorado output.

S13 Focal stack and local-surface construction

For a focal pixel p , the stack is the slice through the focal spatial coordinates across all centers. Each element retains its center origin. The stack can be represented as a raster over center coordinates when centers form a regular tile, as an edge list when centers are irregular, or as a distribution for summary. The first two preserve geography. A histogram alone does not.

For the Colorado analysis, each center origin was transformed to signed local coordinates $(\Delta x, \Delta y)$ relative to the focal pixel. The resulting local surface $L_p(\Delta x, \Delta y)$ contains the same synchrony values as the focal stack but preserves their two-dimensional arrangement around the focal pixel. Pair symmetry implies that one canonical edge contributes to both endpoint surfaces, with opposite displacement and a bearing difference of approximately 180 degrees.

The primary stack dataset uses dimensions **center**, **y**, **x**. Center coordinates include integer row and column indices and source coordinate values. The dense form is intended for bounded scientific inspection. It is not the scalable storage plan for a national all-pairs calculation.

S14 Spatial indexing and geometry

Each pixel is assigned a flat index from its row and column. Geometry is calculated between center coordinates and every focal coordinate. Geographic latitude-longitude inputs use great-circle distance and forward bearing. Projected inputs use Euclidean distance only when their coordinate contract and units support that interpretation. The implementation does not infer a geographic coordinate system from plausible numeric ranges.

The center-to-focal bearing is directional. Reversing pair endpoints changes the bearing by approximately 180 degrees even though the synchrony coefficient is symmetric. Pair attributes must therefore distinguish symmetric edge values from endpoint-specific orientation.

S15 Distance calculations

Great-circle distance is stored in kilometers. A maximum-radius mask is applied before pair calculation, so inactive directed relationships do not generate canonical pair work. Distance-band codes are generated from explicit increasing edges. The default public edges are 25, 50, 100, and 250 km, but the bounded diagnostics used finer radii for stabilization. The Colorado analysis retained 25/50/75/100-km cumulative summaries as diagnostics and added 5-km annular profiles. The 100-km maximum is the observation window, not an inferred scale.

S16 Direction and bearing calculations

Forward bearing is measured from center to focal pixel in degrees. Bearings are classified into N, NE, E, SE, S, SW, W, and NW. Direction is undefined for a self-pair. Direction summaries include group mean and valid count. Unequal group sizes, incomplete neighborhoods, and irregular state-boundary clipping can bias comparisons, so direction should be calculated before focal outputs are clipped to Colorado and interpreted alongside coverage.

For the Colorado surface representation test, angular profiles were calculated after removing the fine radial expectation. This radially adjusted profile asks whether bearing retains information after distance structure is represented.

S17 Nested radius analysis

For radii $R_1 < \dots < R_K$, the diagnostic stack at focal pixel p is restricted to centers satisfying $d(c, p) \leq R_k$. At every radius the workflow records count, median, and IQR. It then calculates each summary's total range, maximum adjacent-radius change, and maximum absolute derivative. These curves show whether a statistic is still responding to newly admitted centers.

When radii are not supplied explicitly, the implementation starts at 0 km, rounds the maximum available distance upward to the next 5 km, and evaluates the full sequence in 5-km increments.

The provisional stable radius is the earliest radius with at least 25 valid centers for which the range over that radius and all larger tested radii is at most 0.02, with at least three radii remaining. This

definition favors a simple auditable rule. It is not an estimator with a sampling distribution.

Among 64 complete Colorado surfaces, the experimental scale screen returned 45 candidate plateaus, 16 multiscale cases, and three right-censored or unresolved cases within the 100-km window. These are diagnostic labels rather than accepted characteristic scales.

S18 Empirical synchrony-range method

The empirical-range pilot asked a different question from the nested-radius diagnostic. The nested diagnostic asks whether a selected focal summary changes as cumulative support increases within an observation window. The range pilot asks whether newly encountered pairwise synchrony approaches an empirical distant background and whether the cumulative reduction stabilizes at the same time. A provisional stable radius from the first calculation is therefore not an estimate of R^* from the second.

The pilot reused the real PRISM data from 1 November 2023 through 30 January 2024 and the established cold/warm tail-Spearman pair calculation. Self-pairs were excluded. For each focal pixel, a single canonical pair table out to a 500-km discovery radius supplied the fixed 100-km control, physical-distance annuli, cumulative summaries, background comparisons, adaptive reductions, and sensitivity tests. The primary annulus width was 20 km; 10- and 40-km widths were also tested. No distance weights or parametric decay kernels were applied.

For tail $h \in \{\text{cold}, \text{warm}\}$, annular medians describe the synchrony of newly admitted pairs and cumulative medians describe the reduction over all pairs within each outer radius. The distant background was estimated from outer relationships rather than fixed at zero. The retained analysis compared outer-annulus, smoothed-outer-annulus, and distant-pair alternatives. A candidate R_h^* required both the annular median to be near background and the cumulative median to be stable for three consecutive supported annuli, followed by a later-shell agreement check. A candidate at or beyond 80 percent of discovery support was labeled censored or unresolved.

Cold and warm ranges were estimated separately. Primary adaptive ΔS was defined only when both resolved, using

$$R_{\text{common}}^*(p) = \max\{R_{\text{cold}}^*(p), R_{\text{warm}}^*(p)\}. \quad (\text{S11})$$

Cold, warm, and pairwise ΔS were then reduced over the identical nonself neighbor set within R_{common}^* . The adaptive contrast was $\text{median}[S_{\text{cold}}(i, j) - S_{\text{warm}}(i, j)]$, not a difference between marginal medians calculated over separate ranges. If either tail failed to resolve, common range and adaptive ΔS remained missing. The discovery boundary was never substituted as an estimated range.

The output retained the tail-specific and common status, boundary-censor flag, later-shell agreement, and reliability score. Reliability could not override an unresolved status. Discovery-radius sensitivity used 200, 300, 400, and 500 km with the same 500-km pair product. The spatial evaluation used five national representative pixels and 25 spatially balanced Colorado sites with full CONUS land support. The retained walkthrough does not report the exact numerical near-background, cumulative-stability, minimum-annulus-count, or reliability tolerances. Those values must be recovered from the pilot configuration before the method is independently reproducible.

EMPIRICAL RANGE PROVENANCE PLACEHOLDER: Insert the exact focal-site coordinates and selection rule; minimum supported-bin count; near-background, cumulative-stability, later-shell, and reliability tolerances; status-code definitions; pair counts by site; run fingerprint; software environment; and machine-readable sensitivity outputs.

S19 Empirical synchrony-decay method

The decay pilot preserved the hard-horizon experiment and used the same retained pair tables to ask which curve properties remained measurable without a finite endpoint. For each tail h , near-field synchrony $S_{\text{local},h}$ was the robust median of the first two well-supported physical annuli. The three retained empirical-background definitions were the outer-annulus, smoothed-outer, and distant-pair alternatives used in the range pilot. Self-pairs remained excluded.

For annular median synchrony $S_h(d)$, normalized excess was

$$E_h(d) = \frac{S_h(d) - S_{\text{bg},h}}{S_{\text{local},h} - S_{\text{bg},h}}. \quad (\text{S12})$$

It is approximately one near the focal pixel and approaches zero at the empirical background. Curves with insufficient or nonpositive local excess retained explicit status flags.

Fractional distance d_q is the interpolated physical distance at which fraction q of local excess has decayed. Thus d_{25} retains 75 percent of local excess, d_{50} retains 50 percent, and d_{75} retains 25 percent. A centered three-annulus median and persistent later agreement prevented one noisy annulus from defining a crossing. Fractions not reached within discovery support remained unresolved rather than being assigned 500 km.

Effective synchrony length was

$$L_h = \int_0^{D_{\text{max}}} \max\{E_h(d), 0\} dd. \quad (\text{S13})$$

It is the width of a height-one rectangle with the same positive area under the normalized-excess curve. It has units of kilometers but is not a hard cutoff, kernel bandwidth, or dispersal distance. Initial slope β_h was the Theil–Sen median slope of supported annular medians in the first 100 km, reported per 100 km [Sen, 1968]. Unlike d_q and L , it does not require an empirical background.

All metrics were estimated independently for cold and warm synchrony. Cold- minus-warm differences in d_{50} , L , or β describe contrasts in curve shape or spatial scale; none is ΔS . Candidate metrics had to meet coverage, representative-site, discovery-radius, bin-width, and background- sensitivity criteria for both tails. Discovery support and annulus-width sensitivity retained 200/300/400/500 km and 10/20/40 km, respectively. The retained report does not provide every numerical smoothing, crossing, fit-quality, or decision tolerance, so these remain archival requirements.

DECAY PILOT PROVENANCE PLACEHOLDER: Insert exact definitions for well-supported annuli, centered smoothing, crossing persistence, fit quality, background tolerances, discovery and bin-width stability, local-versus-far slope comparison, site-level outputs, run fingerprint, and software environment.

S20 Stack statistics

The reduction operation calculates valid-center count, spatial coverage, mean, median, standard deviation, IQR, median absolute deviation, extrema, selected quantiles, an exponentially distance-weighted mean, and near-field median. It also calculates distance-band medians and counts and compass-direction means and counts. Main-text claims use statistics that were directly inspected against the full center-location raster.

Coverage is defined relative to requested centers, not to an abstract complete circle. For statewide work, a second denominator should record the number of geometrically eligible centers within the radius. This separates missing or invalid pairs from unavailable support at a computational boundary.

S21 Local-surface representation comparison

The Colorado analysis compared six representations against the complete retained ΔS surfaces: median plus IQR, cumulative summaries at 25/50/75/100 km, a 20-value fine radial profile, a 43-value radial-plus-directional profile, a seven-value low-order two-dimensional basis, and an eight-score empirical PCA/SVD basis. Reconstruction error was summarized with the report-defined normalized RMSE for complete surfaces and after withholding 10 percent of surface values. The retained report does not state the normalization denominator, so the archival methods must recover that definition from the implementation before these values are independently reproducible.

Normalized RMSE for complete surfaces was 1.01 for median plus IQR, 1.02 for the cumulative-radius baseline, 0.98 for the fine radial profile, 0.74 for the radial-plus-directional profile, 0.81 for the low-order basis, and 0.82 for the empirical basis. Corresponding 10-percent-withheld errors were 1.01, 1.02, 0.99, 0.77, 0.81, and 0.81. Eight empirical components retained 79 percent of training variance, but held-out normalized RMSE remained 0.82 and split-half subspace stability was 0.31. These diagnostics support transparent radial and directional projections while leaving both low-order and empirical bases experimental. Complete cold, warm, and ΔS surfaces remain audit objects. The 64 retained surfaces form a diagnostic coverage sample, not a probability sample for statewide inference.

S22 Radius stabilization diagnostics

The bounded low-IQR radius example was focal (17, 5), with ΔS IQR approximately 0.056. Its median and IQR stabilized near 15 km under the provisional rule. The high-IQR focal (1, 5) had IQR 0.242 and continued changing until approximately 80 km. Across the block, median stable radii ranged 15–80 km with median 35 km; IQR stable radii ranged 15–100 km with median 40 km. The observed ranges of the radius-dependent median statistic were approximately 0.027–0.282, and IQR ranges were approximately 0.056–0.311. Pixels with no qualifying stable radius should remain missing rather than being assigned the largest radius automatically.

S23 Empirical synchrony-range pilot results

The adaptive-range analysis did not meet its prespecified criteria at the primary 500-km discovery radius and 20-km annulus width. Among the 25 Colorado pilot sites, nine resolved a cold range and nine resolved a warm range, but only one site resolved both and could form R_{common}^* . Common-range coverage was therefore 4 percent; 24 of 25 estimates, or 96 percent, were censored or unresolved. None of five national representative pixels resolved a common range. No Colorado pilot site stabilized between 400 and 500 km under the primary bin width.

The prespecified criteria required at least 70 percent common-range resolution, at least 70 percent discovery-stable estimates, and resolution for at least four of five national representative pixels. Observed values were 4 percent, 0 percent, and zero of five. The evidence therefore did not support a full Colorado or CONUS adaptive-range analysis. Approximately 1.18 million nonself pairs were evaluated in the 25-site pilot.

The single resolved common-range site was insufficient to estimate map-level Pearson or Spearman agreement, RMSE, sign-change frequency, or threshold- exceedance rates between adaptive and fixed ΔS . Tail-specific ranges remain descriptive outputs, but their different spatial resolution patterns prevent them from being substituted for a common-neighbor contrast. The pilot likewise did not support an association between common range and fixed-radius ΔS IQR or the reduction gap. These relationships retain an insufficient-support status rather than a null interpretation.

Where both tail ranges resolve, $\Delta R = R_{\text{cold}}^* - R_{\text{warm}}^*$ describes a difference in spatial extent. It is not ΔS , which describes a difference in synchrony strength. With only one common-range site, the pilot did not interpret the geography of ΔR .

Sensitivity across 10-, 20-, and 40-km bins and 200-, 300-, 400-, and 500-km discovery limits showed substantial censoring and boundary dependence. This does not establish that synchrony lacks a finite range. It establishes that the predeclared scalar criterion was not identifiable or stable often enough in this one winter interval to justify adaptive maps. The fixed 100-km control and its previously reported values were unchanged.

S24 Empirical synchrony-decay pilot results

The decay pilot used the same 25 Colorado sites and five national representative pixels. At the primary 500-km discovery radius and 20-km annulus width, d_{50} resolved for cold and warm at all 25 Colorado sites and all five representative sites. Effective length was calculable at all sites. Initial slope passed its fit-quality requirement at 96 percent of cold sites and 84 percent of warm sites. These coverage results show that curve characteristics can be available even when a finite common horizon is not.

The complete decision criteria also required stability. Cold and warm d_{50} were stable from 400 to 500 km at 76 and 52 percent of Colorado sites and across 10/20/40-km bins at 80 and 88 percent. Effective length was discovery stable at only 12 and 16 percent, while its bin-width stability was 76 and 92 percent. Initial slope was discovery stable at all supported sites, but bin-width stability was 40 percent for cold and 84 percent for warm. d_{50} and L met the background-sensitivity tolerance at all 25 sites under the three retained background definitions; β is background independent by construction.

No candidate met all criteria for both tails. Warm d_{50} failed the 70-percent discovery-stability requirement, effective length was strongly discovery-boundary dependent, and cold β failed the 70-percent bin-width requirement. None of d_{50} , L , or β therefore supported a full Colorado or CONUS mapped product.

The metrics nevertheless sharpened interpretation of the fixed control. The median fraction of local excess synchrony lost by 100 km was 0.35 for cold and 0.25 for warm. Near-field decline was steeper than the far-field tail at 84 percent of cold sites and 56 percent of warm sites. These descriptive results are consistent with a relatively fast local component and a broad slower tail in this window. They do not establish a universal two-scale model.

S25 Directional diagnostics

For a grouping variable g , descriptive eta squared is

$$\eta_g^2 = \frac{\sum_h n_h (\bar{x}_h - \bar{x})^2}{\sum_k (x_k - \bar{x})^2}, \quad (\text{S14})$$

where h indexes distance bins or direction groups. The high-IQR example had $\eta^2 = 0.104$ for distance and $\eta^2 = 0.290$ for direction, while its distance-residual IQR remained 0.185. Across the block, distance eta squared had median approximately 0.10 and maximum approximately 0.45. Direction eta squared had median approximately 0.285 and maximum approximately 0.629. Because the groupings overlap in geographic meaning, these numbers cannot be added or interpreted as unique causal contributions.

Preferred follow-up models include smooth distance effects with cyclic bearing terms, spatially varying coefficients, and explicit topographic covariates. Those analyses are not part of the validated bounded result.

At focal (1, 5), cold synchrony had median 0.787 and IQR 0.218, warm synchrony had median 0.825 and IQR 0.175, and pairwise ΔS had median -0.063 and IQR 0.242. Both component fields varied. The broader cold field contributed more spread to the contrast, but neither component can be discarded.

Part IV

Supporting and exploratory spatial diagnostics

S26 Panel-change metrics

For adjacent center landscapes A and B , finite shared pixels define vectors a and b . The retained metrics include:

$$Q = \text{cor}[\text{rank}(a), \text{rank}(b)], \quad (\text{S15})$$

$$\text{RMSE} = \sqrt{\text{mean}(a - b)^2}, \quad (\text{S16})$$

$$\text{NRMSE} = \frac{\text{RMSE}}{Q_{0.95}(a \cup b) - Q_{0.05}(a \cup b)}. \quad (\text{S17})$$

Thus normalized RMSE uses the pooled robust 5th-to-95th percentile range. Sign disagreement compares deadbanded classes $-1, 0, 1$. Near-tie fraction is the fraction of panel values whose nearest sorted neighbor is within a chosen epsilon, 0.005 in the default diagnostic.

Spatial gradient change is computed from finite-difference gradients in the two raster directions. One metric compares gradient vectors; a second compares gradient magnitudes. These metrics require a regular focal raster and are not equivalent to map correlation.

S27 Why rank similarity alone was insufficient

The selected adjacent ΔS landscapes had $Q = 0.088$ but $\text{RMSE} = 0.080$. Their narrow ranges and dense near-ties made ranks unstable. Normalized RMSE of 0.429 and sign disagreement of 0.535 showed that the panels also differed in substantive ways. Warm-panel Q was approximately 0.986 and cold-panel Q approximately 0.590. The diagnostic therefore supports a cold-driven reorganization in this comparison, not a claim that all pairwise differences were large.

Focal heterogeneity and moving-center landscape change answer different questions. The former summarizes variation among relationships incident on one focal location; the latter summarizes how a complete center-reference field changes when the center moves. Their correspondence was modest in the bounded experiment, so neither can be substituted for the other.

S28 Exploratory multimodality methods

Gaussian kernel density estimates are evaluated at 0.75, 1.0, and 1.25 times Scott's bandwidth [Scott, 1992]. Modes must exceed a 5 percent density-prominence rule. Consensus multimodality is recorded only when all three bandwidths retain at least two modes. The multi-bandwidth screen reflects the sensitivity of apparent modes to kernel smoothing [Silverman, 1981]. A minimum group size of 20 applies to the descriptive two-group field. The screen should be rerun with explicit self-pair handling and coverage thresholds in the Colorado analysis.

The procedure is a screen, not formal evidence for a mixture distribution. It does not estimate the number of climate regimes, compare generative models, or correct for spatial multiple testing.

S29 Spatial coherence of candidate groups

Candidate groups are mapped to their contributing center coordinates. A candidate is scientifically more interpretable when groups form coherent geographic areas than when labels are interspersed. Coherence remains descriptive because spatial autocorrelation is expected among nearby centers and reduces the effective independence of nominal observations [Dutilleul et al., 1993]. The final analysis should compare candidate maps with edge distance, coverage, and support-radius curves before interpretation.

Twenty-two of 400 bounded-experiment focal pixels passed the bandwidth-consensus screen. Many occurred near domain margins. Interior focal (9, 13) retained two modes and coherent center groups. This evidence warrants follow-up and does not establish discrete climate regions.

S30 Exploratory relational-convolution feasibility study

A separate feasibility report examined whether local relationship fields could support convolution-like comparisons under relative-offset and absolute- geographic alignment. It was an exploratory prototype rather than part of the primary analysis. The report recorded a ΔS relationship correlation of 0.71 under the relative formulation, an absolute-geographic correlation of 0.94, absolute-gradient similarity of 0.93, and much weaker relative-gradient similarity near 0.18. Its preliminary station-level ΔS correlation was 0.68. Synthetic recovery and practical-window checks were encouraging, but the analysis was not promoted to a public analysis method.

The feasibility report also listed radial-plus-directional and empirical-basis errors of 0.68 and 0.55, whereas the reviewed 64-surface representation analysis reported 0.74 and 0.82. These values are not combined because the retained reports do not establish that they used the same target, sample, masking, or normalization. The discrepancy must be resolved from code and configuration before any relational-convolution result is cited as confirmatory evidence. Accordingly, this study remains an exploratory repository artifact and does not support a main-text claim.

Part V

Numerical validation and sensitivity

S31 Bounded pairwise and surface validation

The validated block included 400 centers, 160,000 directed relationship slots, and 80,200 canonical pairs. The recorded run required 71.024 seconds on one process, processed approximately 1,129 pairs per second, reached 23.2 MB peak traced memory, and produced a 5.0 MB in-memory xarray object. Center-map equivalence had maximum absolute error 0.0 for cold, warm, and ΔS .

The audited focal pair was 44.8378 km from the reference center. It returned $S_{\text{cold}} = 0.588180125$ with 40 joint observations and $S_{\text{warm}} = 0.857692301$ with 39 joint observations. The contrast was $\Delta S = -0.269512177$. Manual average-rank calculations agreed with the package result to 10^{-7} .

S32 Bounded sensitivity and diagnostic analyses

Phase 1.5 reused the Phase 1 checkpoint identified by analysis fingerprint

```
sha256:4b4a3c30fbd5ccb2a4f8b5cefc2ca2b127e7bf31d646b27091b354b447df9acb
```

It added panel-change, radius, distance, direction, KDE, and geographic-coherence diagnostics without recomputing the pair correlations.

The repository base commit was `a852137944cabf238e9321f2eecf09e483c9b9f5`. The report qualifies the analysis as that base commit plus uncommitted Phase 1/1.5 changes.

Key sensitivity findings were: (1) similar median ΔS could coexist with IQR near 0.242 versus 0.062; (2) a low panel Q required complementary magnitude, sign, and near-tie diagnostics; (3) stable radii varied across the block; (4) the high-IQR example was directionally organized but remained bandwidth-stable unimodal; and (5) focal heterogeneity and adjacent-landscape change had only modest correspondence.

Part VI

Computational reproducibility and statewide execution

S33 Software and implementation provenance

The climate-synchrony analysis uses five operations implemented in CubeDynamics [Tuff and Contributors, 2026]:

- `local_synchrony_stack`;
- `reduce_synchrony_stack`;
- `stack_radius_diagnostics`;
- `stack_structure_diagnostics`; and
- `panel_change_diagnostics`.

These operations are implementation details rather than the scientific subject of the technical report. The stack calculation reuses the validated single-tail kernel and records split inequalities,

minimum counts, window semantics, geometry, pair symmetry, batch size, and an analysis fingerprint.

The implementation represents labeled arrays with xarray [Hoyer and Hamman, 2017] and uses Dask for deferred and parallel blocked computation [Rocklin, 2015]. Exact package versions remain required in the archived environment.

The primary source is `src/cubedynamics/`; legacy mirrors and prose are not implementation truth. The final archive must pin the exact commit, package version, environment lock, configuration, and input manifests.

S34 Tiling and halo architecture

The Colorado analysis used center tiles with focal and data halos at least as large as the 100-km observation window. Canonical pairs were assigned deterministically so overlaps did not duplicate final edge records. Reduced focal outputs were emitted only for tile interiors, while external centers remained available during calculation. Political boundaries were applied after neighborhood calculation.

The real 100 by 100 pixel benchmark compared tile-halo execution, endpoint reversal, chunking, reconstructed surface values, and reduced signatures. Maximum absolute errors were at or below numerical precision. This benchmark passed before the Colorado summaries and landscape-change branch were executed.

S35 Computational scaling

For N pixels in a dense unrestricted domain, unique undirected pairs grow as $N(N+1)/2$. A finite maximum radius changes the practical scaling toward the number of local edges, but density, grid resolution, and radius remain decisive. The project record explicitly rejects dense national all-pairs computation.

The 100 by 100 benchmark evaluated 3,254,392 canonical pairs in 32.4 seconds with peak resident memory of 1.16 GB. Colorado focal summaries evaluated 24,922,476 pairs in 248.1 seconds with peak resident memory of 1.21 GB. The separate landscape-change calculation evaluated 26,412,407 pairs in 359.6 seconds with peak resident memory of 1.23 GB. Retaining 64 complete focal surfaces required 12.6 seconds for approximately 120,000 pairs and stored 14.8 MB. These are run-specific observations, not package-wide performance claims. The archival record must still identify hardware, task counts, chunk sizes, compression, checkpoint frequency, and restart behavior.

S36 Reproducibility and provenance

Every stack records a SHA-256 analysis fingerprint constructed from schema version, variable names, window dates, minimum count, split quantile, grid shape, center indices, coordinate hashes, maximum radius, and distance bands. Checkpoints require both NetCDF data and a complete manifest with a matching fingerprint. Incompatible resume is rejected.

The final provenance bundle should additionally include source checksums, retrieval records, CubeDynamics commit, dependency lock, run configuration, tile manifests, logs, validation comparisons, figure-building scripts, and hashes of exported figure data. A semantic trace assists inspection but is not complete workflow provenance.

The two retained copies of the empirical-range walkthrough are byte identical. Their SHA-256 hash is `5341712bedc1683996107c207370019125909067e0ab98cb70b0dd7994fbce88`. This identifies the reviewed report but does not replace the missing immutable configuration, machine-readable outputs, or source-code commit for the pilot.

The retained empirical-decay walkthrough has SHA-256 hash `eec05135003ee60f1789dd485330b220002cdd6810bcaa88ad8ea816b47ce7ce`. It likewise identifies the reviewed report rather than a complete reproducible analysis bundle.

S37 Generative artificial intelligence assistance and oversight

Generative artificial intelligence was used for technical-report preparation and repository editing, not as a source of empirical evidence. The disclosure in the main technical report is expanded here so that the system identity, workflow, task boundaries, and verification steps are explicit.

Table S1: Detailed record of generative-AI assistance during technical-report preparation.

Element	Recorded information
Provider and interface	OpenAI Codex accessed through the Prism research-writing environment.
Model identification	The environment identified the agent as based on GPT-5. It did not expose a more specific model snapshot or API model string, weights or build identifier, temperature, top-p, random seed, or serving revision. No more specific version designation is inferred.
Dates of documented use	24–25 September 2026.
Interaction mode	Iterative natural-language instructions with workspace read/write access, explicit file patches, a sandboxed command line for searches and document builds, and public web search for bibliographic verification.
Materials supplied	The evolving main technical report, Supplementary Information, BibTeX database, analysis reports, figure and supplementary-figure plans, reviewer and citation audits, repository documentation, existing figures, and conceptual-figure source code.
Principal instructions	Read the complete project record; reorganize the main paper around scientific findings rather than analysis chronology; retain negative results in the scientific record; move validation, sensitivity, and implementation detail to the supplement; preserve the cold-minus-warm sign convention; keep historically referenced tails as future work; identify claims needing citations; and do not invent analyses, results, parameters, or archive identifiers.
Editorial tasks	Structural editing of the main paper and supplement; line-level rewriting; terminology and sign-consistency checks; placement of methods, results, limitations, and future work; caption and title refinement; and preparation of audit and repository documentation.

Table S1: Detailed record of generative-AI assistance during technical-report preparation (continued).

Element	Recorded information
Citation tasks	Identification of potentially unsupported claims and missing literature areas; searches of primary and publisher records; preparation of proposed BibTeX entries; insertion of selected citations; and scripted checks for missing, unused, or duplicate keys and DOI values.
Code and figure tasks	Assistance with the repository-native Python script that renders the conceptual hierarchy and tile-and-halo schematic. These are explicitly conceptual figures. Empirical maps and numerical figure values came from the supplied analysis products and reports.
Mechanical verification	Searches across LaTeX and project files, comparison of claims with supplied reports, citation-key and DOI audits, and repeated LaTeX/BibTeX compilation of the technical report and supplement.
Excluded uses	The model did not collect or generate PRISM or GHCN-Daily data, replace the CubeDynamics pairwise calculations, run the primary climate analysis, select undocumented scientific parameters, create missing empirical results, infer unrecorded provenance, or serve as an author.
Human oversight	Members of the ASymClim Working Group supplied the scientific objectives and claim boundaries, iteratively corrected and redirected the model, selected which changes to retain, and retain responsibility for all scientific claims, citations, interpretations, figures, code, and final text.
Reproducibility limit	Because the exact serving snapshot and stochastic settings were not exposed, the language-model interaction cannot be reproduced exactly from the provider and date alone. The final archive should retain this disclosure and, where institutional and provider policies permit, the prompt and edit history needed to reconstruct the editorial process.

S38 Colorado analysis details

The bounded 100 by 100 pixel benchmark passed its numerical checks, after which the statewide focal domain was the Colorado state polygon and the source and center domains extended through the 100-km local observation window. The analysis produced cold median, warm median, pairwise ΔS median, ΔS IQR, ΔS median absolute deviation, and separate landscape-change products. Cumulative 25/50/75/100-km summaries were retained as diagnostics rather than treated as the final surface representation.

COLORADO PARAMETER PLACEHOLDER: Insert exact state boundary source and version, geodesic buffering method, rasterization rule, focal-pixel inclusion rule, halo distance, tile dimensions, pair batch size, date window, split quantile, minimum tail count, coverage threshold, self-pair rule, and output schema.

COLORADO ARCHIVE VALIDATION PLACEHOLDER: Insert exact error values for tile-boundary equality, endpoint reversal, chunking, surface reconstruction, and signature comparison; add pair-count conservation, missingness, coverage, range, checksums, run fingerprint, and restart-test results.

Part VII

Observational comparison, interpretation, and limitations

S39 Exploratory GHCN-Daily station comparison

An exploratory quality-control analysis compared PRISM relationships with raw Global Historical Climatology Network-Daily observations [Menne et al., 2012]. Fifty-four of 60 station candidates met a 70-percent completeness screen, and 30 were selected spatially. These stations generated 35 eligible edges no longer than 150 km. Nonblank GHCN quality flags were excluded. Same-label comparisons were primary, with plus-or-minus-one-day sensitivity retained because PRISM and station day definitions can differ.

Station and PRISM thresholds were calculated within their respective data sources. For cold-tail pair synchrony, Pearson agreement across the 35 edges was 0.84, Spearman rank agreement was 0.87, mean absolute error was 0.11, and RMSE was 0.14. For warm-tail synchrony the corresponding values were 0.03, 0.30, 0.12, and 0.20. For ΔS they were 0.68, 0.62, 0.15, and 0.21.

This is not an independent validation. Edges share stations, the graph is sparse, PRISM is station informed, and the contribution status of each selected station to the exact PRISM product-period roster was unknown. The contrasting cold and warm results nevertheless identify station-level and spatial-block holdouts as a required future validation step.

S40 Supplementary ecological interpretation

The analysis characterizes the environmental covariance field and does not measure population synchrony, dispersal, demographic rates, trophic interactions, or biological response functions. It therefore does not test the Moran effect directly. Spatial variation in environmental autocorrelation can, however, alter how shared forcing is translated into population synchrony [Massie et al., 2015], and warming can alter both environmental and population synchrony [Hansen et al., 2020]. Local dynamics can amplify, dampen, or transform that forcing [Liebhold et al., 2004, Vasseur and Fox, 2009]. The local synchrony surfaces provide an environmental template that can be joined to ecological time series in future analyses.

Environmental relationship heterogeneity is not biological response diversity [Elmqvist et al., 2003]. A broad ΔS surface does not measure resilience, and ΔS IQR is not a response-diversity index. This distinction is consistent with broader response-diversity theory [Mori et al., 2013]. Spatial insurance arises when components do not fluctuate in perfect synchrony [Yachi and Loreau, 1999, Loreau et al., 2003]; related portfolio effects have been demonstrated in exploited populations [Schindler et al., 2010]. Their ecological expression depends on biological responses and interactions as well as environmental covariance [Loreau and de Mazancourt, 2008]. These connections motivate coupled analyses; they are not results of the present temperature-only study.

S41 Limitations

The current empirical evidence represents one winter window. Relative median splits are not absolute extremes. Conditional correlations can be sensitive to joint sample size and threshold ties. PRISM values are physiographically interpolated observational surfaces [Daly et al., 2008]. Spatial autocorrelation reduces the effective independence of pair and pixel summaries [Dutilleul et al., 1993]. Direction and distance diagnostics are descriptive. Stable radii and KDE modes depend on tuning choices. The experiment does not measure biological synchrony, response diversity, resilience, causal mechanisms, or climate regions.

The statewide extension reduces the original block-boundary limitation but does not establish national or long-term generality. The 64 complete surfaces are a sparse representation sample, and the empirical basis was not stable across splits. Repeated windows, seasons, source products, quantile choices, station comparisons, and biological data are required for broader claims.

The adaptive-range pilot adds a separate limitation. It sampled 25 Colorado sites and five national representative pixels, not a statewide or national raster. Its result establishes that the specified criteria were not met; it does not establish infinite synchrony range. Exact background and convergence tolerances, site coordinates, pair counts by site, and status-code definitions still require an immutable pilot archive.

The decay pilot reused the same small site sample. High metric coverage does not imply stability, and the reported proportions are not statewide estimates. d_{50} , L , and β depend differently on discovery support, annulus width, background, smoothing, crossing, and fit-quality choices. None passed the complete criteria for both tails, and no mapped decay product was generated.

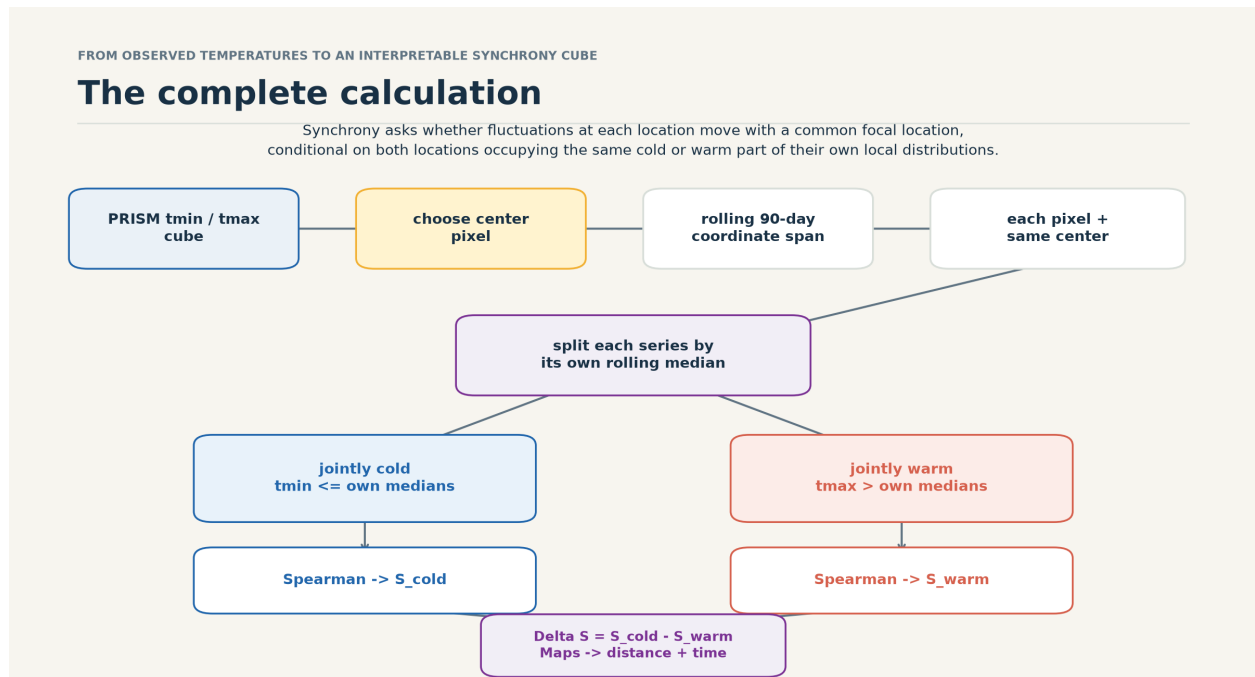


Figure S2: Implemented climate-synchrony calculation from PRISM minimum and maximum temperature through the rolling window, pixel-specific median splits, joint-tail masks, Spearman coefficients, and the cold-minus-warm contrast. Canonical pair symmetry is applied when the center-reference calculations are assembled into focal stacks.

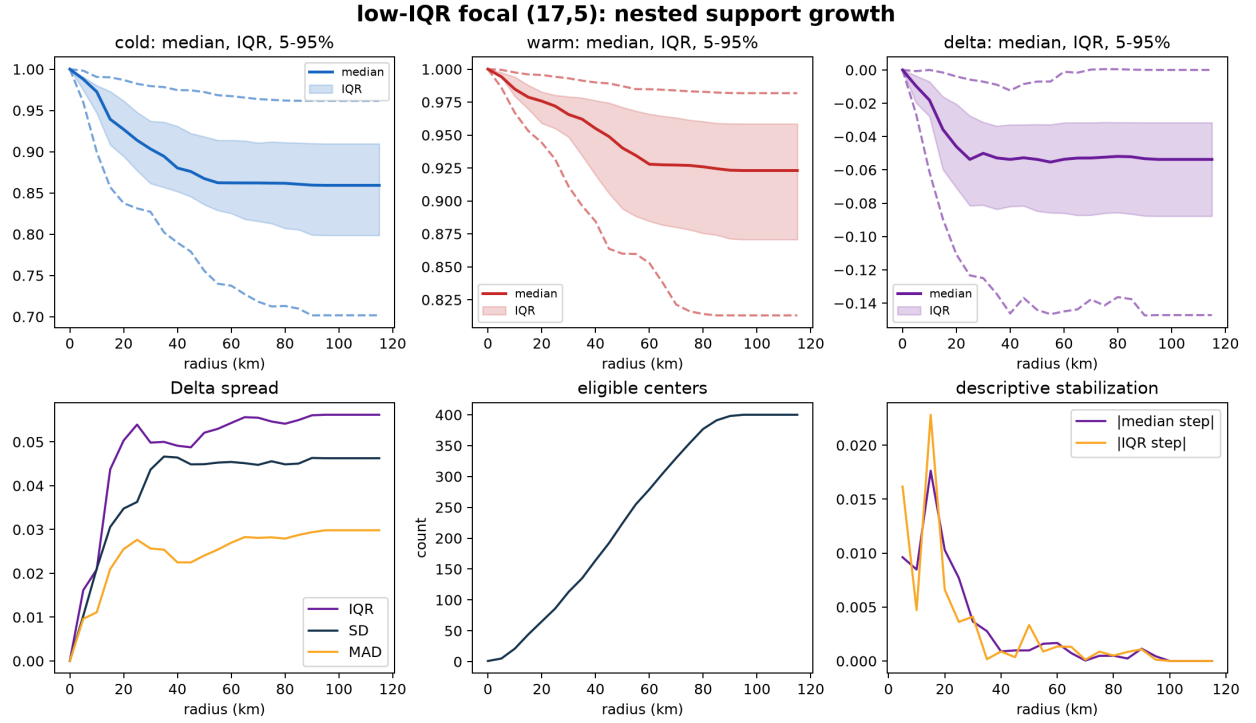


Figure S3: Nested support growth for the low-IQR radius example at focal (17, 5). Panels retain separate cold, warm, and ΔS medians and spreads, eligible-center count, and stepwise diagnostic changes. Both provisional stable radii were 15 km for this bounded window.

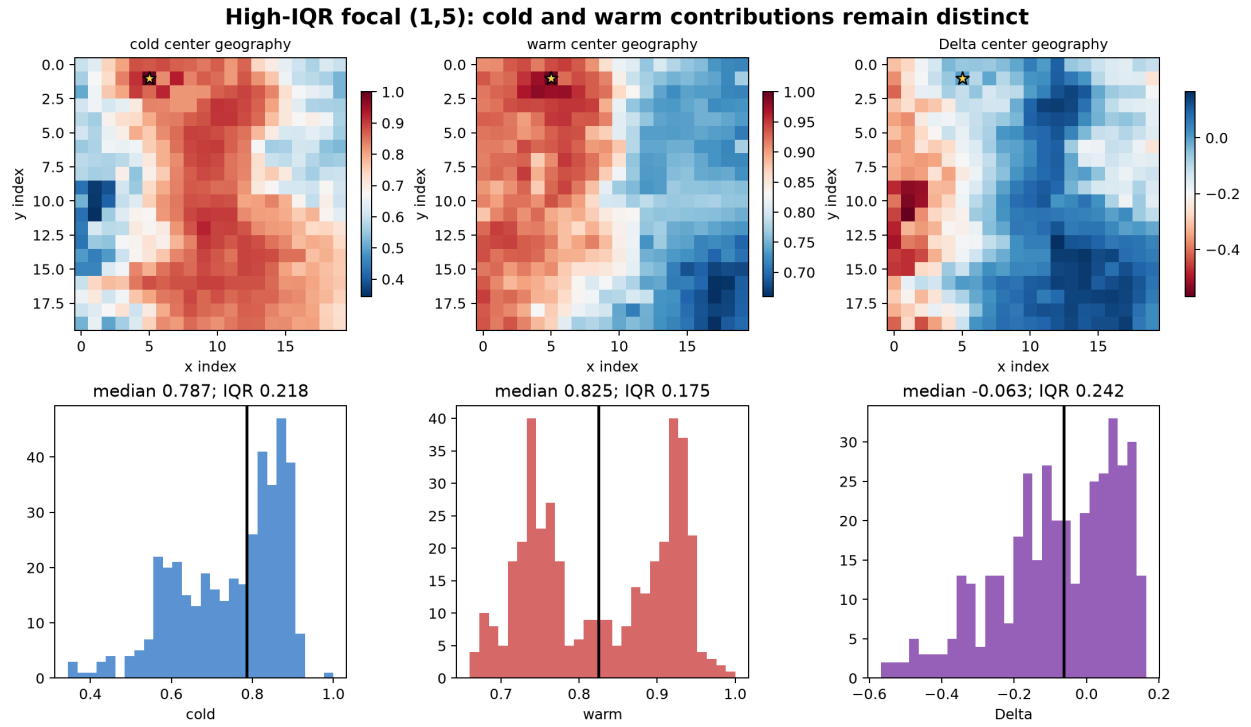


Figure S4: Cold, warm, and pairwise ΔS center geography and distributions for high-IQR focal (1,5). The bottom row reports the component medians and IQRs retained in the scientific interpretation.

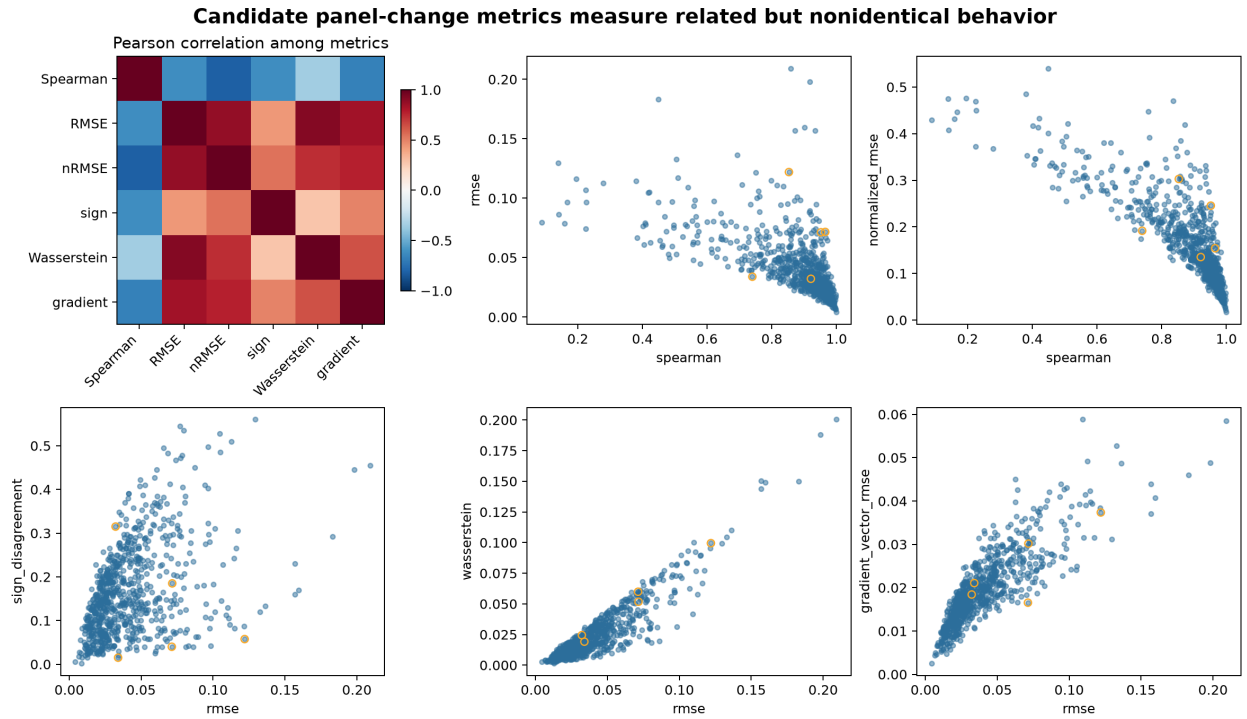


Figure S5: Comparison among candidate adjacent-landscape change metrics. Spearman similarity, cosine similarity, Pearson similarity, RMSE, normalized RMSE, sign disagreement, and gradient diagnostics preserve different aspects of panel change. Wasserstein distance was strongly redundant with RMSE in this block and did not preserve geography.

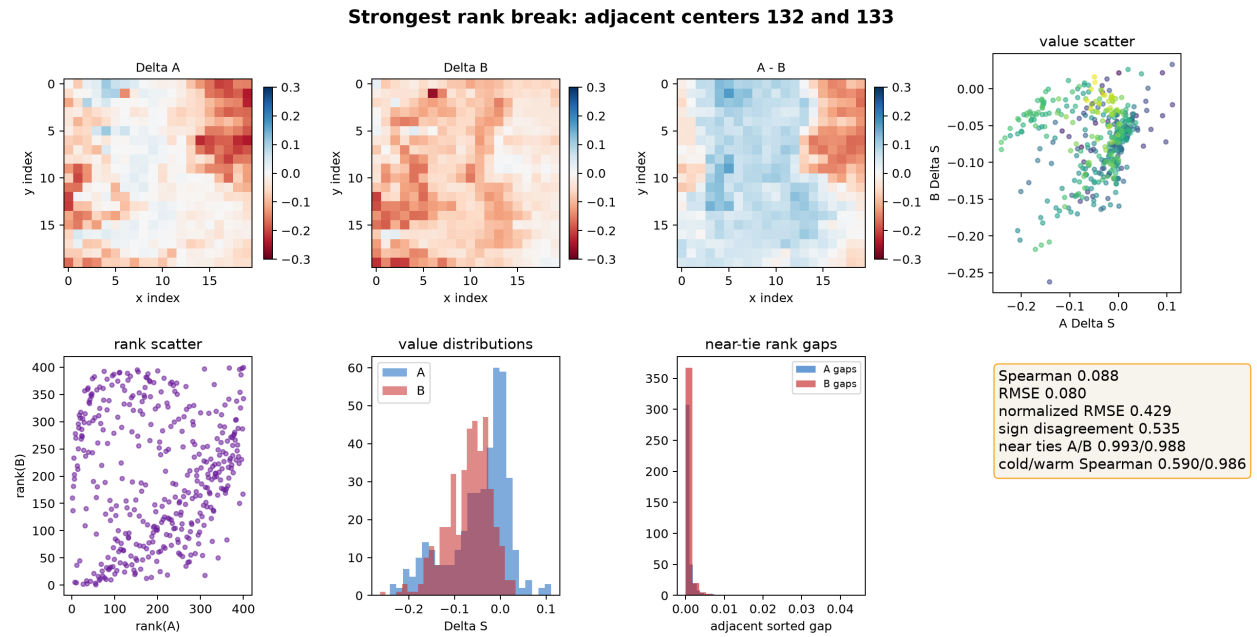


Figure S6: Full diagnostic view of the adjacent center-reference landscapes with the lowest initial ΔS rank similarity. The panels show both landscapes, their pixelwise difference, value and rank comparisons, deadbanded sign change, and cold–warm decomposition. This figure supports the conclusion that landscape change is multidimensional and that the selected example was driven primarily by reorganization of the cold component.

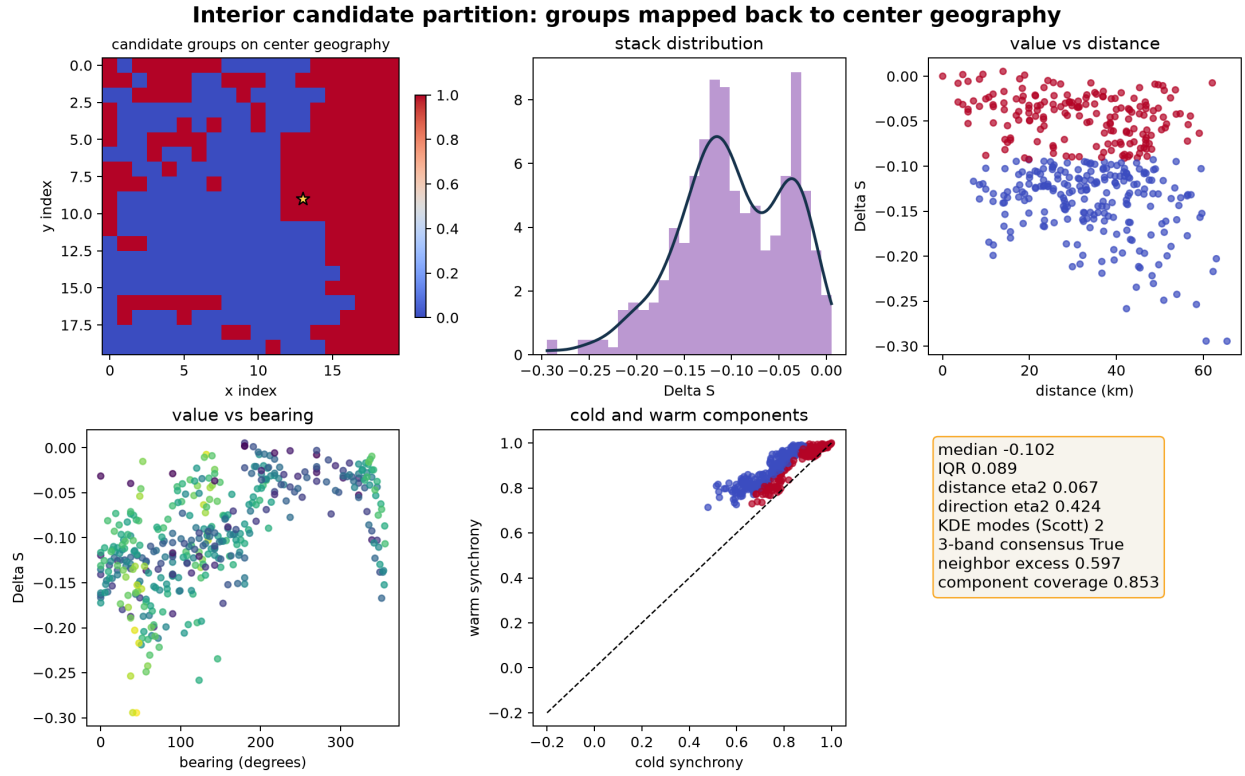


Figure S7: Exploratory candidate spatial partition at interior focal (9,13). Two KDE modes persisted across 0.75, 1.0, and 1.25 times Scott's bandwidth, and forced two-group labels occupied coherent center geography. The screen is not a climate-region classification.

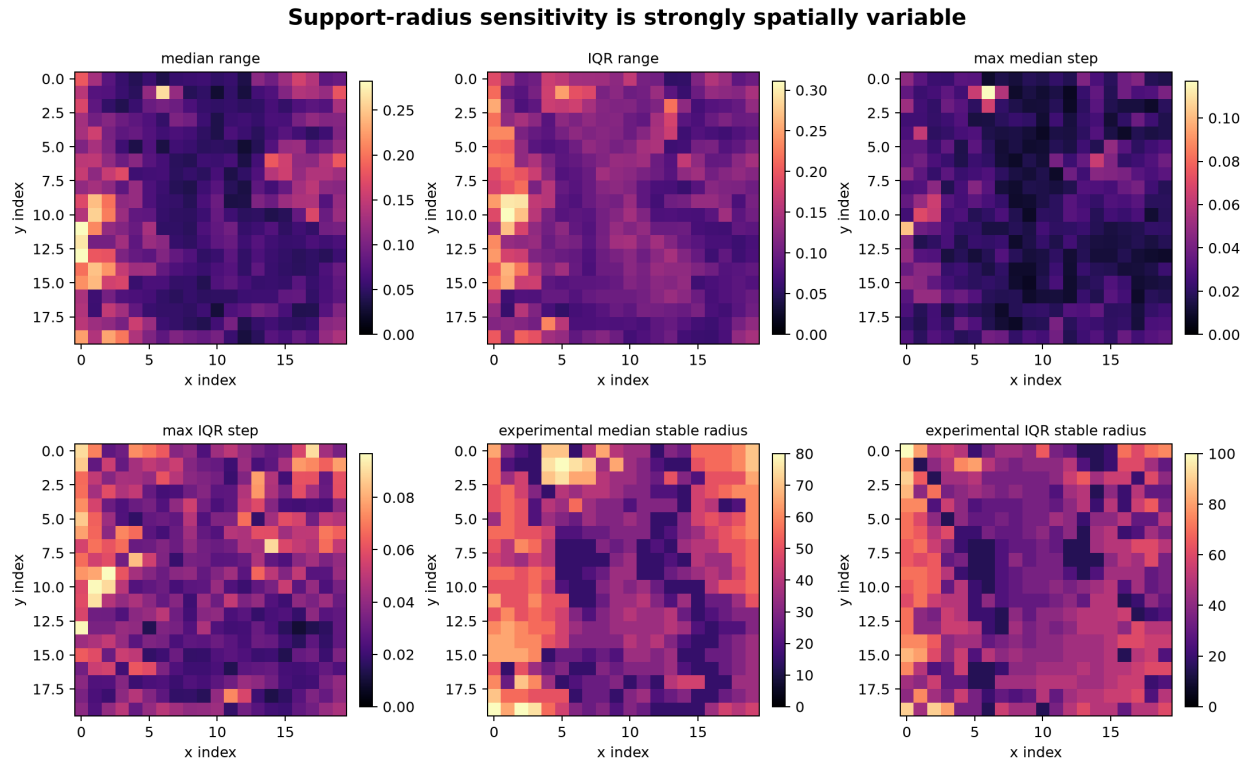


Figure S8: Spatial variation in radius sensitivity across the 20 by 20 block. Maps show median and IQR ranges, maximum steps, and the experimental stable radii. These diagnostic scales are spatially variable and threshold dependent.

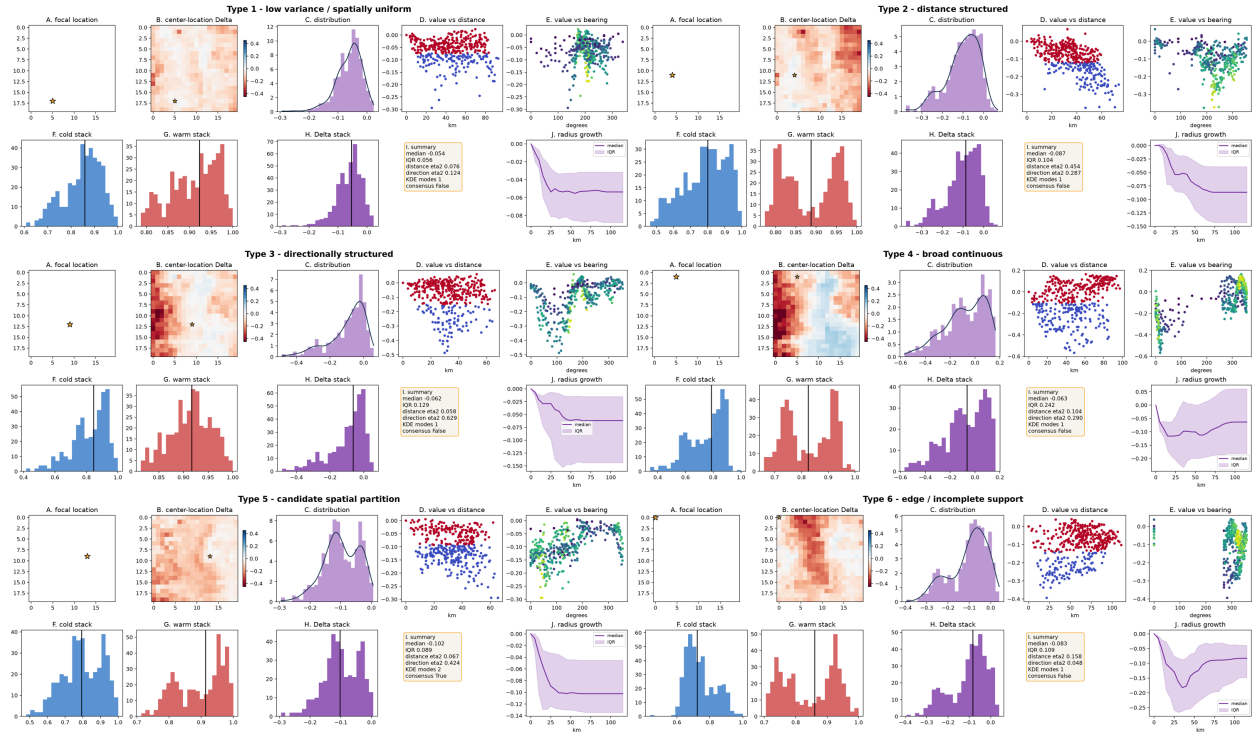
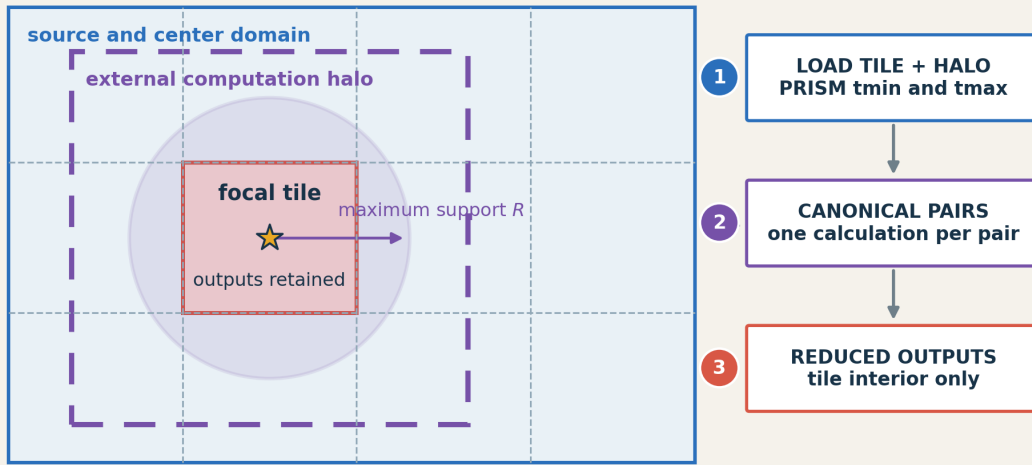


Figure S9: Six descriptive stack-selection roles observed in the block: low variance, distance structured, directionally structured, broad continuous, candidate spatial partition, and edge or incomplete support. These are examples selected for interpretation, not inferred climate classes.

FINITE-RADIUS TILING WITH AN EXTERNAL SCIENTIFIC HALO

Compute neighborhoods first; clip focal outputs only after validation



Conceptual architecture; tile size and final halo remain Phase 2 parameters

Figure S10: Conceptual tile and halo architecture for the Colorado analysis. Focal outputs are retained only for a tile interior, while source data and eligible centers are loaded across an external scientific halo. Canonical pairs are calculated once across overlaps. The 100 by 100 benchmark established boundary equivalence before statewide execution.

Part VIII

Reference tables

S42 Supplementary tables

Table S2: PRISM variables and temporal coverage. Retrieval details still require the archived Colorado manifest.

Variable	Role	Bounded window	Colorado coverage
tmin	Cold-tail synchrony	2023-11-01 to 2024-01-30	Same window
tmax	Warm-tail synchrony	2023-11-01 to 2024-01-30	Same window

Table S3: Synchrony parameters for the validated experiment and Colorado analysis.

Parameter	Bounded experiment	Colorado analysis
Split quantile	0.5	0.5
Nominal window length	90 days, inclusive	90 days, inclusive
Complete daily labels	91	91
Minimum joint tail count	10 in audited example	Archive manifest required
Spatial supports	5-km diagnostic sequence	100-km window; 5-km annuli
Self-pair handling	Included	Excluded in fixed control; archive per product
Direction classes	8	Radially adjusted angular profile
Panel adjacency	4-neighbor	4-neighbor
Sign deadband	0.02	0.02, plus sensitivity
Near-tie epsilon	0.005	0.005, plus sensitivity

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Table S4: Empirical synchrony-range pilot and prespecified scale-up criteria. Resolution counts refer to the primary 500-km discovery radius and 20-km annulus width.

Diagnostic	Required	Observed
Colorado cold range resolved	Not a criterion alone	9 of 25
Colorado warm range resolved	Not a criterion alone	9 of 25
Colorado common range resolved	At least 70%	1 of 25 (4%)
Colorado discovery-stable	At least 70%	0 of 25 (0%)
National representative common range	At least 4 of 5	0 of 5
Stabilized between 400 and 500 km	Not a criterion alone	0 of 25
Mapped-product outcome	Meet all criteria	Criteria not met

Table S5: Empirical synchrony-decay pilot coverage and stability at the primary 500-km discovery radius and 20-km annulus width. Percentages refer to the 25 Colorado pilot sites unless representative-site counts are shown.

Diagnostic	Cold	Warm
d_{50} resolved	100%	100%
d_{50} representative sites	5 of 5	5 of 5
d_{50} discovery stable	76%	52%
d_{50} bin-width stable	80%	88%
L calculable	100%	100%
L discovery stable	12%	16%
L bin-width stable	76%	92%
β fit quality passed	96%	84%
β discovery stable	100%	100%
β bin-width stable	40%	84%
Complete-criteria outcome	Not met	Not met

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Table S6: Bounded pairwise and surface-validation statistics.

Statistic	Value
Center landscapes	400
Directed center-to-focal relationships	160,000
Canonical pairs including self-pairs	80,200
Runtime	71.024 s
Throughput	1,129 pairs s ⁻¹
Peak traced memory	23.2 MB
In-memory stack size	5.0 MB
Maximum cold map error	0.0
Maximum warm map error	0.0
Maximum ΔS map error	0.0

Table S7: Bounded local-surface examples. The median-matched low-IQR example and the radius-growth low-IQR example are different focal pixels.

Example	Median ΔS	IQR	Approx. stable radius	Interpretation
Focal (1, 5)	-0.063	0.242	80 km	Broad, directional
Focal (8, 14)	-0.062	0.062	Not reported	Median matched
Focal (17, 5)	Not reported	0.056	15 km	Radius example
Median-matched difference	0.0015 before display rounding			

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Table S8: Six descriptive selection roles used to inspect stack structure. They are examples, not inferred climate classes.

Focal	IQR	Distance η^2	Direction η^2	Modes	Role
(17, 5)	0.056	0.076	0.124	1	Low variance
(11, 4)	0.104	0.454	0.287	1	Distance structured
(12, 9)	0.129	0.058	0.629	1	Direction structured
(1, 5)	0.242	0.104	0.290	1	Broad continuous
(9, 13)	0.089	0.067	0.424	2	Candidate partition
(0, 0)	0.109	0.158	0.048	1	Edge/incomplete

Table S9: Selected adjacent-panel change metrics and their interpretation.

Metric	Approximate value	Interpretation
Delta rank similarity Q	0.088	Strong rank reordering
RMSE	0.080	Absolute magnitude difference
Normalized RMSE	0.429	Difference relative to range
Sign disagreement	0.535	Different deadbanded sign
Warm Q	0.986	Warm ranks largely retained
Cold Q	0.590	Greater cold reorganization

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Ty Tuff and CubeDynamics Contributors. Cubedynamics, 2026. URL <https://github.com/CU-ESIIL/cubedynamics>. Analysis should cite the exact commit used for the final archived run.

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Table S10: Candidate synchrony-signature variables. These are a menu of distinct properties, not a fixed composite index.

Property	Candidate variables
Magnitude	Cold median, warm median, ΔS median
Heterogeneity	IQR, MAD, standard deviation, quantile span
Support	Fine annular profile, scale-screen diagnostics
Distance	Fine radial profile, smooth decay parameters
Direction	Radially adjusted angular profile, half-plane contrast
Coverage	Valid count, eligible count, valid fraction
Distribution shape	Bandwidth-consensus mode screen, skewness if justified
Residual structure	Low-order two-dimensional reconstruction error
Landscape change	RMSE, rank, sign, gradient, cold/warm decomposition

Table S11: Colorado workflow computational results.

Run	Wall time	Canonical pairs	Peak RSS
100 by 100 benchmark	32.4 s	3,254,392	1.16 GB
Colorado summaries	248.1 s	24,922,476	1.21 GB
Landscape change	359.6 s	26,412,407	1.23 GB
64 complete surfaces	12.6 s	approximately 120,000	14.8 MB stored

Table S12: Colorado comparison of compact local-surface representations. Lower normalized RMSE indicates better reconstruction.

Representation	Parameters	Full RMSE	10% missing	Status
Median plus IQR	2	1.01	1.01	Diagnostic
25/50/75/100 km	8	1.02	1.02	Diagnostic
Fine radial profile	20	0.98	0.99	Retain
Radial plus directional	43	0.74	0.77	Retain
Low-order 2-D basis	7	0.81	0.81	Experimental
Empirical PCA/SVD	8	0.82	0.81	Experimental

Table S13: Recorded software version and environment fields still required from the archived run.

Software	Version
CubeDynamics	0.1.0rc3 in retained records; base commit <code>a8521379</code> plus changes
Python	Pending archived environment
xarray	Pending archived environment
NumPy	Pending archived environment
Dask	Pending archived environment
SciPy	Pending archived environment

Table S14: Bounded benchmark and Colorado run parameters.

Parameter	Value
Bounded validation domain	100 by 100 pixels
Primary focal domain	Colorado
External halo	100-km observation-window support
Support representation	5-km radial; adjusted angular; cumulative diagnostics
Climate variables	PRISM tmin , tmax
Split quantile	0.5
Minimum tail count	Pending archived manifest
Dates/window	2023-11-01 to 2024-01-30
Canonical summary pairs	24,922,476
Coverage threshold	Pending archived manifest
Reconstruction nRMSE denominator	Pending source and manifest recovery

Table S15: End-to-end cold, warm, and ΔS semantics audit.

Audit element	Result
Tail assignment	Lower local tmin is cold; upper local tmax is warm
Pair algebra	$\Delta S = S_{\text{cold}} - S_{\text{warm}}$
Five real-pair recomputations	Maximum absolute error 2.22×10^{-16}
Known-answer controls	Cold stronger +0.925; warm stronger -0.925; equal 0
Primary focal reduction	Median of pairwise ΔS ; not difference of medians
One-pixel storage trace	NetCDF error 0; GeoTIFF error 2.55×10^{-9}
CONUS one-window summary	Median ΔS -0.017335; 70.5% negative
Technical-report color contract	Red negative/warm; white zero; blue positive/cold
Terminology	“Warm” canonical; “hot” only a legacy alias